Raman Ebrahimi

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Summary

Quantitative Analyst with a strong mathematical foundation and expertise in AI and quantitative analysis. I possess strong analytical and programming skills, with experience in optimization, probabilistic modeling, and data analysis. Passionate about leveraging computational techniques, optimization, and statistical analysis, I aim to uncover insights, drive impactful solutions, and contribute to innovative projects by solving complex real-world problems with cutting-edge quantitative methods.

Skills

- **Programming Languages:** Python, SQL, JavaScript
- **Other Tools & Technologies:** Tableau, Gurobi, PowerBI, AnyLogic, NetLogo, LaTeX
- **Research Skills:** Mathematical modeling, Game Theory, Optimization, Critical thinking, Problem-solving

Experience

- **Founding Quantitative Researcher** — *May 2025 - present*
  MarketCrunch AI [] — Remote, USA
  - Researched and enhanced **predictive models** for **2100+ individual stocks and market indicators**, leveraging **mathematical and statistical techniques**. Conducted **exploratory data analysis (EDA)** to identify patterns and inform model improvements which resulted in **a consistent 3% improvement in prediction accuracy** and **6% improvement in directional accuracy**. Developed and backtested **algorithmic trading strategies** for **market prediction** and **portfolio optimization** using Python and quantitative finance libraries. Decreased the individual prediction wait time by **35%** by restructuring the prediction pipeline.
- **Data Science Intern** — *April 2025 - present*
  HCM Tradeseal [] — Remote, USA
  - **Optimized Python ETL pipelines** to extract, clean, and transform data from diverse sources into **SQL databases**, improving data integrity. Built **data validation frameworks** to detect anomalies and missing values. Automated **data quality workflows** using scripting and rule-based systems. Applied **regex, NLP, and statistical modeling** to enhance parsing logic and pattern recognition in **unstructured/semi-structured data**. Streamlined data pipeline automation for scalable validation and database management. Designed CI/CD pipelines to have scheduled and change based deployment to update and analyze data. The final product has **380,000+ searchable data points** extracted and analyzed from various sources over the web.
- **Graduate Student Researcher** — *Sep 2022 - June 2025*
  Multi-agent Intelligence and Decision Systems (MINDS) lab [] — San Diego, USA
  - Developed **game-theoretic frameworks** and **AI/ML models incorporating cognitive biases** using **mathematical optimization** and Python simulations. Analyzed **Nash equilibrium** conditions in multilayer network games to **derive policy insights** from real-world data. Designed **interactive strategy games** with ML-driven simulations to study human-environment interactions and strategic behavior. Combined theoretical modeling with computational techniques to bridge game theory and practical decision-making applications.
- **Data Analyst Intern** — *Dec 2024 - Feb 2025*
  American Institute for Behavioral Research and Technology [] — San Diego, USA
  - **Revived a Python/SQL-based graphing application** for the "Generativity Theory" project, restoring functionality after years of inactivity. **Integrated ML (PyTorch, scikit-learn)** for user-agnostic behavior prediction, achieving **>90% accuracy** in select game modes. Enhanced the model via **mathematical optimization** and **operations research** techniques. Built **data recording & visualization tools (pandas, matplotlib, plotly)** to streamline behavioral analysis and cross-subject validation.
- **Data Science Product Manager** — *Jan 2019 - Oct 2021*
  Scientific Association of Industrial Engineering [] — Tehran, Iran
  - Led a team of CS students in **developing a real-time strategy game** in Unity, overseeing **risk management**, **storyline design**, and **technical execution**. Designed in-game economics, performed **market prediction**, and ran simulations to test servers and identify exploits. **Monitored live data** during competitions to detect anomalies and adjust policies. Produced an aggregated report analyzing 1,000+ data points to improve future events.

Education

- **University of California, San Diego** — *Sep 2023 - Mar 2028 (expected)*
  *Ph.D. in Machine Learning and Data Science (ECE department) - **Advisor:** Massimo Franceschetti* — San Diego, USA
- **University of California, San Diego** — *Sep 2023 - Mar 2025*
  M.Sc. in Machine Learning and Data Science (ECE department) — San Diego, USA
  - **Thesis:** From Games to Predictions: Strategic Behavior in Networks and Classification
- **Sharif University of Technology** — *Sep 2017 - Aug 2022*
  B.Sc. in Industrial Engineering (3.8/4.0) and B.Sc. in Physics (3.9/4.0) — Tehran, Iran

PROJECTS

- **SendSense iOS app: Implementation of the performance prediction tool in an iOS application.**
  Tools: Swift, eXplainable AI (XAI), Machine Learning, Data Analysis []
  - iOS app for athletes to use to log training sessions and utilize machine learning to optimize their performance.
  - Implemented explainable AI to suggest actionable recourse.
  - Created interactive charts using Swift, ensuring insightful visualizations.
- **Algorithmic Trading with Reinforcement Learning**
  Tools: Python, TensorFlow, Gym, Backtrader
  - Designed and implemented a Deep Reinforcement Learning trading agent using Proximal Policy Optimization (PPO) to optimize trade execution and portfolio allocation.
  - Built a simulated trading environment using historical stock/crypto data and compared RL performance with momentum and mean-reversion strategies.
- **Order Book Analysis & Market Microstructure Modeling**
  Tools: Python, Pandas, Scikit-learn, XGBoost
  - Collected and analyzed Level-2 market data from Binance API to model bid-ask spreads, order flow imbalances, and short-term price movements.
  - Developed a model using LSTMs and XGBoost to forecast asset price changes based on order book dynamics.

PUBLICATIONS

C=Conference, J=Journal, S=In Submission

[S.1] Matthew Chan, Steve Sharp, Jiajian Zhu, **Raman Ebrahimi**. **Layers of a City: Network-Based Insights into San Diego's Transportation Ecosystem**. Manuscript submitted for publication in *TRB Annual Meeting 2026*.

[C.4] **Raman Ebrahimi**, Kristen Vaccaro, and Parinaz Naghizadeh. **The double-edged sword of behavioral responses in strategic classification: Theory and user studies**. Manuscript submitted for publication in *ACM Conference on Fairness, Accountability, and Transparency (FAccT 2025)*.

[J.1] **R. Ebrahimi** and P. Naghizadeh, **United We Fall: On the Nash Equilibria of Multiplex and Multilayer Network Games,**. In *IEEE Transactions on Control of Network Systems*.

[C.3] **Raman Ebrahimi**, Kristen Vaccaro, and Parinaz Naghizadeh. **The Double-Edged Sword of Behavioral Responses in Strategic Classification.** ***(Spotlight talk)*** *NeurIPS 2024 Workshop on Behavioral Machine Learning*.

[C.2] **Ebrahimi, R.**, Naghizadeh, P. (2025). **Extended Horizons: Multi-hop Awareness in Network Games**. In: Sinha, A., Fu, J., Zhu, Q., Zhang, T. (eds) *Decision and Game Theory for Security. GameSec 2024*. Lecture Notes in Computer Science, vol 14908. Springer, Cham.

[C.1] **R. Ebrahimi** and P. Naghizadeh, **United We Fall: On the Nash Equilibria of Multiplex Network Games.**, *2023 59th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, Monticello, IL, USA, 2023, pp. 1-8.

HONORS AND AWARDS

- **Awarded a $40,000 scholarship for Master of Quantitative Finance program** *April 2025*
  University of California, San Diego
- **Reviewer for *American Control Conference*, *Automatica*, and *TRB***

REFERENCES

Massimo Franceschetti
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